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Frontier language models violate payoff-scale invariance and show stronger persistence than reciprocity in the prisoner's dilemma across languages

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Large language models increasingly act as autonomous agents negotiating and deciding for human users, making their cooperative dispositions a central question for AI governance. We examine how two fundamental levers shape cooperation: the nominal magnitude of the stakes and the language in which the interaction is framed. In classical game theory, multiplying all payoffs by a positive constant is an invariant transformation that leaves preference rankings, best replies, and equilibria untouched. Across five frontier models evaluating an iterated prisoner's dilemma across ten payoff scales and five languages, this invariance is systematically violated. Rescaling stakes alters cooperation rates dramatically, driving divergent shifts across models that cancel out when averaged. Scale sensitivity also varies across languages. Payoff scale also changes persona compliance: at sub-unit stakes, all five models cooperate more under the selfish than the cooperative persona; in Claude and Gemini 3.5 this effect reverses sign above the sub-unit regime. Across the evaluated models, own-action conditioning is larger than partner-action conditioning. Because scale sensitivity is already visible in opening moves, cooperation metrics cannot be treated as intrinsic model traits without specifying payoff scale and language.

1. Introduction

Large language models are increasingly deployed as *agents* that negotiate, coordinate and decide on behalf of their users, in recommendation systems, negotiation tools and multi-agent assistants [1–4]. In these settings behaviour emerges from strategic interaction rather than from isolated model outputs, and the agents meet *cooperation dilemmas* whose resolution bears directly on safety, coordination and AI governance [5–8]. A fast-growing literature measures how they behave when they do, with the instruments of behavioural economics and cognitive psychology [9–11] and with the classical games [12]. Equally consequential is how far that behaviour adapts to the costs and benefits at issue, since a deployed agent meets the same dilemma at many different magnitudes [3].

A growing body of work evaluates language models through game-theoretic lenses, and in particular through matrix and repeated games, reporting systematic departures from equilibrium play, persistent cooperative biases and sensitivity to contextual factors such as language and incentives [13–20]. The prisoner's dilemma is the workhorse of that literature, its strategy space mapped and its predictions sharp [21–26]. Our study is closest to FAIRGAME, which introduced the reusable game-theoretic evaluation framework and reported payoff and language sensitivity in repeated games [13]. We retain its no-communication prisoner's dilemma protocol so that the comparison is anchored to the same task, while adding a positive-rescaling null spanning ten scales, five languages and six model endpoints. This makes the contribution a test of invariance and heterogeneity, rather than a new game or a new payoff definition.

Within it the models turn out to be individual rather than uniform, differing in how readily they retaliate [27], how readily they forgive [28], how much weight they give the game rather than the story around it [29] and how far an instructed disposition carries into conditioned reciprocity [30]. These studies share two unremarked reporting conventions. Each summarises a model by a cooperation rate, or a few of them, measured at one payoff matrix; and each reports what the agents did rather than the decision rule that produced it.

Evaluating strategic behaviour at the level that alignment and governance require therefore asks for more than aggregate outcomes. Drawing on behavioural and evolutionary game theory [31], a behavioural intention can be conceptualised as an agent's *strategy*: a decision rule mapping an interaction history to the next action [32–35]. The canonical strategies of repeated games, Always Cooperate, Always Defect, Tit-for-Tat and Win-Stay-Lose-Shift [31, 36], supply an interpretable vocabulary for such intentions, and prior work has shown that they can be inferred from noisy behavioural trajectories by supervised learning [32, 33] and extended by probabilistic models to mixed or stochastic play [37]. The inference is not free. Separating the memory-one rules from observed play has been shown to need a long run of observations even where the resident rule is fixed by construction [38, 39], which is why the read-out used here is reported as a resemblance, with its provenance beside it (§2.4).

In parallel, what a language model does is not invariant across the language it is addressed in [13]. Strategic reasoning, risk sensitivity and cooperation patterns depend on linguistic framing, sometimes by margins comparable to the differences between models [29, 40, 41]. Whether the strategies agents play are themselves language-dependent, and whether that dependence introduces systematic bias into multi-agent settings, is largely open.

That dependence is one instance of a wider pattern: what is measured on a language model depends on things the measurement was not about. Accuracy moves with the order of the in-context examples [42], and with formatting choices that leave the meaning of the prompt intact, by margins wide enough to swamp the differences between the models being compared [43]. Instructions about who the model is are equally consequential: a persona changes what a model produces [44], shifts measured capability [45] and, conditioned on a demographic backstory, reproduces the response distribution of the matching human subgroup [46]. What none of these manipulations can supply is a null.

The choice of units is not usually stated as a design decision at all, because in the theory it is not one. A payoff matrix is a von Neumann-Morgenstern utility, defined only up to a positive affine transformation [47, 48], so writing the same dilemma with cells ten times larger changes no preference, no best reply, no equilibrium and no orbit of the replicator dynamics. Under the theory the two matrices are not two games written differently. They are one game. The theory of the iterated dilemma inherits the indifference, turning on the ordering of the four payoffs and on ratios among them and never on their absolute size [21, 23, 24, 26], and the modelling literature normalises the scale away [49–51]. Current theory works at the indifference rather than merely permitting it: where social learners update on the payoff they last received, the condition for conditional cooperation to persist under strong selection loses the benefit and the cost of cooperation altogether and turns on the ordering of the payoffs alone [52].

The invariance belongs to the theory, not to people: human valuation is neither linear in magnitude nor anchored at zero [53], human choice reverses between logically equivalent descriptions of one problem [54], and human cooperation in the dilemma itself responds to the size of the stakes [55, 56]. That gap has been argued to be structural rather than incidental, with the linguistic description of an action entering the utility function directly and a shift proposed from outcome-based to language-based preferences [57].

Two things are established, and they have not been put together. The first is that what a language model does depends on features of the prompt that carry no task content: the order of the examples, the formatting, the persona, the language. The second is that multiplying every payoff by a positive constant carries no task content in the strictest sense available, because the matrix it produces is not a second game but the same one. What is not known is whether the strategic behaviour of a language model respects that invariance. To our knowledge, no prior study has formulated positive payoff rescaling as a theorem-backed zero-effect invariance test across a broad multi-model, multilingual grid. Every manipulation in the sensitivity literature could legitimately change behaviour, so a measured change under any of them is at worst a nuisance and at best a finding, against an expected effect nobody had specified. A rescaling is the one manipulation whose expected effect is specified, and specified as zero, so a measured departure has a stated reference value rather than a comparison condition.

What follows if the invariance fails is a question about what a published cooperation rate means. Machine behaviour proposes that autonomous systems be studied empirically, with the instruments of the behavioural sciences, because what such a system does is not deducible from how it was built [58]. Those instruments arrive with a known difficulty: a behavioural manipulation almost never carries a predicted effect, so a measured change has to be read against a comparison condition the experimenter chose. The invariance classes of decision theory are the exception, supplying a manipulation whose predicted effect is a stated number, and the number is zero. The consequences run through the strands of machine behaviour this study touches. The persona line is the one place where the experimenter addresses the agent about its social role, and whether it takes hold turns out to depend on the units the matrix was printed in. The departure is already present on the opening move, which locates it in the reading of the matrix rather than in the weighing of a history. And models of mixed human-machine populations carry the cooperativeness of the artificial agent as a parameter and ask where the population settles [59–63]; a rate partly determined by the experimenter's units is not the quantity those models need.

Together, these considerations let us put four questions to the rescaling:

(1) Does strategic behaviour survive multiplying every payoff by a positive constant, and by how much does it fail to?
(2) Does the answer depend on the prompt language, and on the persona the model is told to adopt?
(3) Does the rescaling change only how often a model cooperates, or also the strategy it is playing?
(4) Does the effect enter at the opening move, or through the course of play?

We answer them on a complete and exactly balanced grid of 10,000 dyads (§2), and read the answers against an evolutionary baseline computed on the same grid.

Here, we turn a theorem into an instrument. The contributions of the present paper are: **(i)** a payoff-rescaling protocol whose predicted behavioural effect is exactly zero, demonstrating that five frontier language models systematically violate scale invariance (§3.1), with cooperation rates shifting substantially across stakes. **(ii)** a per-model analysis revealing that scale departures diverge sharply across models to the point of anticorrelation, so that pooled analyses produce an artificial average that describes no real model: variance is dominated by model individuality and model-by-scale interactions. In the five-model cell-mean decomposition, the shared scale main effect is largely carried by Grok, whereas the dyad-level pooled association persists in leave-one-model-out fits. **(iii)** cross-lingual and structural evidence showing that scale sensitivity depends heavily on the prompt language, gates persona compliance by inverting instructions at sub-unit stakes, erodes textbook strategy conformity, and emerges on the opening move before interaction begins. **(iv)** an analytical evolutionary baseline that moves in the opposite direction to the pooled empirical corpus. The transition analysis shows stronger persistence than reciprocity, without identifying a causal mechanism for cooperation (§3.8, appendix D.7).

An earlier and smaller collection on this project has been reported separately by the present authors [64]: three models at three payoff scales, in the same five prompt languages, with cooperation read off the trajectories and reported as rising with the stakes. The present article differs in what it collects and in what it claims. It collects six models rather than three and ten payoff scales rather than three, spanning five orders of magnitude, on a grid whose completeness and balance are enforced by the analysis build rather than assumed. It frames the rescaling as an invariance whose predicted effect is a theorem rather than as a manipulation of the stakes whose effect was expected. And it reports every quantity per model, which is what shows the movement to be heterogeneous, with two of the five running downwards, and the pooled statement to be an artefact of averaging.

2. Material and methods

2.1 Framework

FAIRGAME, the Framework for AI Agents Bias Recognition using Game Theory [13], supplies the computational infrastructure for this study. It supports systematic and reproducible evaluation of language models through controlled game-theoretic experiments: an experimental condition is declared in a configuration file that fixes the payoff structure, the horizon, the model backend and the prompt language; at run time the framework combines that declaration with language-specific prompt templates, simulates the repeated normal-form game, and logs the round-by-round trajectory. We extend it in three ways. First, with a payoff-scaling factor applied to the prisoner's dilemma, which is the manipulation this paper is about (§2.2). Second, with a completeness guard that refuses to release an analysis table unless the grid is exactly balanced and each game's recorded payoffs recover the scale its cell was run at. Third, with a strategy read-out that reports which canonical memory-one rule a trajectory most resembles, and with what provenance (§2.4).

The design is a complete factorial crossing of six language models, ten payoff scales, five prompt languages, four persona pairings and ten replicates. The unit of collection is the dyad, one ten-round game between two agents of the same model, contributing two agent-games, an agent-game being the ten decisions one agent takes in one game. Each of the 300 model-by-language-by-scale cells holds exactly 40 dyads, the four persona pairings by ten replicates; pooling the five languages gives 200 dyads and 400 agent-games in each of the 60 model-by-scale cells. The corpus is 12,000 dyads, 24,000 agent-games and 240,000 decisions. Five of the six models are reported in the main text and the sixth, Gemini 3.1 Flash-Lite, in the appendix, for the reason given with the models below. The five carry 250 of the cells, 10,000 dyads, 20,000 agent-games and 200,000 decisions, and are the grid every main-text figure and table is computed on. Balance is enforced rather than assumed (appendix A.1).

2.2 The payoff-scaled prisoner's dilemma and its invariance

Both agents receive the prisoner's-dilemma prompt of the FAIRGAME protocol [13] in its no-communication, announced-horizon condition. The scenario casts the strategic interaction as a criminal interrogation, states the four payoff outcomes as penalties, names the two available actions Option A and Option B, carries the running history of prior moves and realised penalties, announces a fixed horizon of $T = 10$ rounds, and instructs each agent to minimise its cumulative penalty.

The row player's baseline payoff matrix for the Prisoner's Dilemma is

	Option A	Option B
Option A	$(6, 6)$	$(0, 10)$
Option B	$(10, 0)$	$(2, 2)$

where the first and second entries in each cell denote the penalty payoffs to the row and column player, respectively. In standard game-theoretic notation, this matrix assigns the four canonical outcomes:

$$T = 0, \quad R = 2, \quad P = 6, \quad S = 10,$$

representing the temptation (T), reward (R), punishment (P), and sucker (S) penalty payoffs. The penalty ordering

$$T < R < P < S \quad \text{with} \quad 2R < T + S$$

is the exact isomorphic image of the familiar canonical Prisoner's Dilemma condition $T_{\text{util}} > R_{\text{util}} > P_{\text{util}} > S_{\text{util}}$ and $2R_{\text{util}} > T_{\text{util}} + S_{\text{util}}$ defined on utilities $u = -\text{penalty}$:

$$T_{\text{util}} = 0 > R_{\text{util}} = -2 > P_{\text{util}} = -6 > S_{\text{util}} = -10.$$

Because lower penalties correspond to higher utilities, Option A strictly dominates Option B in every column (facing Option A, choosing A yields penalty 6 versus 10; facing Option B, choosing A yields penalty 0 versus 2). Option A is therefore the defecting action, while mutual Option B yields penalty 2 versus penalty 6 for mutual Option A, making Option B the cooperative action; throughout this paper, cooperation denotes choosing Option B.

The realised score-action relationships reproduce the intended dilemma structure in the logs: own cooperation correlates with own utility at Spearman $\rho = -0.015$, while the opponent's cooperation correlates at $\rho = +0.916$, and dyad welfare correlates with joint cooperation at $\rho = +0.985$. These checks validate the payoff mapping and action polarity, not model comprehension of the dilemma (appendix A.1).

To manipulate the stakes of the game without altering its strategic structure, best-reply correspondences, or Nash equilibria, we introduce a positive scalar parameter $\lambda > 0$ and multiply all four baseline cells by λ (figure 1b). In our experiments, we evaluate ten values spanning five orders of magnitude:

$$\lambda \in \{0.01,\ 0.10,\ 0.25,\ 0.50,\ 1.0,\ 2.0,\ 5.0,\ 10.0,\ 100.0,\ 1000.0\},$$

corresponding to attenuated ($\lambda < 1.0$), baseline ($\lambda = 1.0$), and amplified ($\lambda > 1.0$) payoff regimes. For example, at the sub-unit scale $\lambda = 0.10$, the row player's payoff matrix becomes

	Option A	Option B
Option A	$(0.6, 0.6)$	$(0, 1.0)$
Option B	$(1.0, 0)$	$(0.2, 0.2)$

where every printed payoff is at most 1.0. At the amplified scale $\lambda = 10.0$, it becomes

	Option A	Option B
Option A	$(60, 60)$	$(0, 100)$
Option B	$(100, 0)$	$(20, 20)$

with the ordering $T < R < P < S$ strictly preserved in all cases. The complete matrix for all ten scales is tabulated in appendix A.3.

Under classical decision theory and game theory, multiplying utilities by $\lambda > 0$ constitutes a positive affine transformation of a von Neumann-Morgenstern utility function [47, 48]:

$$u'(a) = \lambda u(a) + c \quad (\lambda > 0,\ c = 0).$$

Such transformations preserve all preference orderings over lotteries, best replies, Nash equilibria, and deterministic replicator dynamics orbits. Consequently, the normative theoretical prediction for the effect of λ on rational agents is exactly zero. This construction isolates the effect of payoff magnitude from structural incentives: any systematic departure from invariance constitutes an empirical discovery about machine behaviour rather than a rational adaptation to changing game-theoretic payoffs.

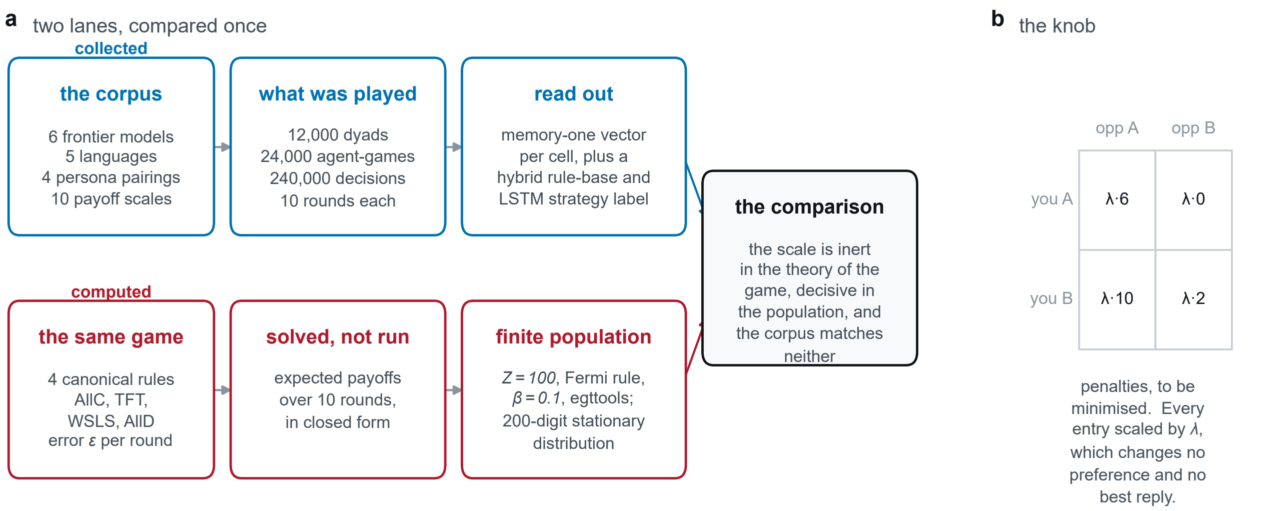


Figure 1. Overview of experimental design, payoff-scaling protocol and analysis lanes. (a) The empirical collection: six frontier language models played against themselves across ten payoff scales $\lambda \in [0.01, 1000]$, five prompt languages, four persona pairings and ten replicates (12,000 dyads, 240,000 decisions). (b) The payoff-rescaling manipulation: multiplying the prisoner's dilemma penalty matrix by λ. Under classical decision theory and deterministic replicator dynamics, multiplying payoffs by a positive constant leaves preferences, best replies and equilibria invariant. (c) The analytical evolutionary lane: solving finite-population imitation dynamics ($Z = 100$, Fermi selection $\beta = 0.1$, execution noise $\epsilon = 0.05$) at 200 significant digits across the same ten payoff scales, yielding the theoretical baseline against which the empirical corpus is compared.

2.3 Models, prompt languages, personas and protocol

Six frontier models were run, each against itself; two different models were never paired. They were Claude Haiku 4.5, GPT-5.4 Nano, Gemini 3.5 Flash-Lite, Qwen3 235B-A22B, Grok 4.20 and Gemini 3.1 Flash-Lite, served through the Kaggle Benchmarks model proxy on an OpenAI-compatible chat-completions endpoint at temperature 1.0. Their endpoint identifiers are listed in appendix A.2. The main text reports the first five. The sixth, Gemini 3.1 Flash-Lite, is reported in full in appendix B.7, and the reason is a property of its endpoint rather than of its results: its identifier is both undated and announced as provisional, so what was served under that name while the corpus was collected cannot be re-addressed afterwards. We state the cost of drawing the line there. The identifier of Gemini 3.5 Flash-Lite, which stays in the main text, is undated as well, so it carries a weaker form of the same risk; the line is between an identifier marked provisional and one that is not, and not between pinned and unpinned. Gemini 3.1 Flash-Lite is also the one model of the six whose strategy trend runs opposite to the rest (§3.5), which is a reason to report it prominently rather than to set it aside. Every quantity the main text gives for the five is given for all six in the appendix, which also states what including it changes, and on every conclusion of this paper that is nothing.

The prompt languages are English, French, Vietnamese, Chinese and Arabic, in templates byte-identical to the FAIRGAME resource files and checked against them before every collection run. Artefacts of those files are reproduced rather than repaired, the consequential one being that the Arabic and Chinese versions state the four cells as penalties but phrase the goal sentence as maximising a reward, so we treat the five languages as five distinct stimuli rather than five renderings of one (appendix A.4). Each agent also receives one line naming its persona, in the language of the prompt: in English, *You are cooperative.* or *You are selfish.* The two are crossed in all four ordered pairings, ten replicates each. The opponent's persona is never disclosed, so a pairing reaches a player only through realised behaviour and not before round 2, and we treat it as a design factor and a regression covariate rather than as information the agent held.

Every game runs ten rounds with simultaneous moves: both agents see the same history, rounds 1 to $r-1$, and the round is scored once both have replied. Sampling uses documented matching by game identifier across payoff scales. The released tables do not retain the API seed values, so this record verifies the matching structure but not provider-specific seed semantics. Decisions are elicited as free text and parsed by the protocol's own matcher. A reply it cannot resolve is regenerated, up to two further attempts, and if all three fail the decision is recorded as Option A, which under this framing is defection: the fallback can only push a measured cooperation rate down, never up, and it cannot manufacture the cooperative behaviour reported here. Each collection run asserts that its fallback rate is at most 0.02, but the achieved rates are not carried in the released per-game tables, so we cannot report how they vary with the payoff scale (appendix A.2).

2.4 Strategy read-out

While the framework provides complete gameplay trajectories and descriptive metrics such as cooperation rates, these capture what agents do rather than the decision rule behind it. Following prior work on inferring canonical strategies from noisy repeated-game data [32–34], we read every agent-game against the four canonical rules AllC, AllD, Tit-for-Tat and Win-Stay-Lose-Shift. The read-out reports which of the four a ten-round trajectory most *resembles*. It is not an identification of the strategy a model is running, and no claim in this paper treats it as one.

All four rules are memory-one, so exact matching is a lookup with no learning in it, and it returns a *set* rather than a label, because over ten rounds several rules routinely prescribe the same moves. The three provenances, the agent-games matched exactly by one rule (deduced), by several at once (ambiguous) and by none (unmatched), are recorded separately and never merged, and the same matcher yields a continuous quantity needing no labels at all, the number of rounds violating the nearest rule. Where the rules leave the answer open a single-layer LSTM [65] arbitrates. Trained on a synthetic corpus alone, which contains no language-model output, it reaches 98.5% accuracy in distribution and 93.6% and 81.1% on two execution-noise levels held out entirely (appendix A.6). It ranks only within the set the rule base returns, so it can never overturn a deduction; it decides alone only where no rule fits, and its share of the labels is very unevenly spread across the four rules, which bounds how the composition may be read (§3.5).

2.5 Statistical analysis

The cooperation rate of an agent-game is the share of its ten decisions that are Option B; a cell mean averages the agent-games of a cell, and round-level analyses use the binary decision itself. Because the two agents of a dyad play each other and the ten rounds inside an agent-game are strongly autocorrelated, primary inferential intervals were computed by resampling whole dyads with replacement, 1,500 draws, percentile method [66], and primary per-model and trend regressions clustered their standard errors on the dyad. The supplementary scale-by-language robustness model instead clusters on the documented language-by-game-ID blocks to preserve the matched ten-scale ladders. Descriptive figure intervals use whole-dyad resampling. Because the range over the ten scales is a maximum minus a minimum, and so positive by construction, its bootstrap interval quantifies the precision of the range and not its distance from zero; the permutation test supplies the test. Because the null is a theorem, the matching test destroys the association with the payoff scale and leaves the rest of the design as observed: the scale label is shuffled among whole dyads, the unit to which the scale was assigned, and the range of the ten scale means recomputed, 2,000 draws, so 0.0005 is the smallest attainable p value.

For each model we fitted a logistic regression of the round-level decision on the payoff scale as a ten-level factor, with prompt language, own persona and opponent persona as covariates, $\lambda = 1$ as reference and standard errors clustered on the dyad, and tested the nine scale contrasts jointly by a Wald test (table 1). The variance decomposition is a type-II analysis of variance on the 250 cell means of the five models reported here, with terms for model, language and payoff scale and their three two-way interactions and no three-way term, so the residual on 144 degrees of freedom is the omitted three-way interaction; the same decomposition on all 300 cells is in appendix B.3. The persona result contrasts the two sub-unit scales with the remaining eight as a difference of differences with an interval from the same dyad bootstrap; where a trend across the whole grid is wanted instead, $\log_{10}\lambda$ enters as a continuous predictor. Analyses were run in Python; the package versions and the seeds are recorded with the archived pipeline in appendix E.

Table 1. Cooperation over the ten payoff scales, by model. Minimum, maximum and range, with a 95% confidence interval on the range from a bootstrap over whole dyads and a permutation test on the payoff-scale label (0.0005 is the floor of 2,000 draws). The last two columns test the nine scale contrasts jointly in a per-model logistic regression of the round-level decision, clustered on the dyad. Each entry pools 200 dyads per scale; ranges are computed before rounding. Full specification in the electronic supplementary material.

	cooperation					Wald test	
Model	min	max	range	95% CI	p_{perm}	χ^2_9	p
Claude Haiku 4.5	0.371	0.452	0.081	[0.062, 0.135]	0.0025	32.3	< 0.001
GPT-5.4 Nano	0.438	0.680	0.243	[0.200, 0.290]	0.0005	164.1	< 0.001
Gemini 3.5 Flash-Lite	0.682	0.872	0.190	[0.165, 0.240]	0.0005	129.0	< 0.001
Qwen3 235B-A22B	0.399	0.494	0.095	[0.068, 0.158]	0.0005	96.9	< 0.001
Grok 4.20	0.274	0.910	0.636	[0.588, 0.686]	0.0005	1227.3	< 0.001

3. Results

Across the 240,000 recorded decisions of the whole corpus, the sixth model included, cooperation is 0.5895; the five models reported below carry 200,000 of those decisions.

3.1 Cooperation moves under a transformation that cannot matter

Under normative decision theory and classical game theory, multiplying every payoff in a normal-form matrix by a positive constant is an axiomatically neutral transformation. It preserves all preference rankings over lotteries, best-reply correspondences, and Nash equilibria: an agent playing the game should not alter its choices whether payoffs are denominated in cents, dollars, or thousands of pounds.

Yet, across five orders of payoff magnitude, every frontier language model evaluated violates this scale invariance (figure 2a,b, table 1). The shift in strategic posture is substantial: across the ten-scale ladder, cooperation swings widely in every model, departing decisively from the invariance null (figure 2c and table 1). What classical game theory treats as an inert change of units produces a fundamental reordering of multi-agent behaviour.

Strikingly, the models do not share a common behavioural archetype: each possesses a distinct machine character that responds idiosyncratically to the scale of the stakes. Grok undergoes a profound behavioural metamorphosis. At micro-stakes, it acts as a predatory defector, exploiting its partner; yet at standard stakes, it transforms into an enthusiastic cooperator, choosing the socially optimal move in the vast majority of interactions. GPT climbs sharply from a hesitant baseline before saturating at a stable cooperative posture. Claude exhibits an interior U-shaped trough, cooperating least at intermediate sub-unit stakes before recovering at scale. Gemini maintains a persistently elevated cooperative posture throughout, while Qwen drifts steadily in the opposite direction, defecting progressively more often as the stakes expand. Because each model traces a unique behavioural curve, machine cooperation cannot be characterised by a single universal archetype.

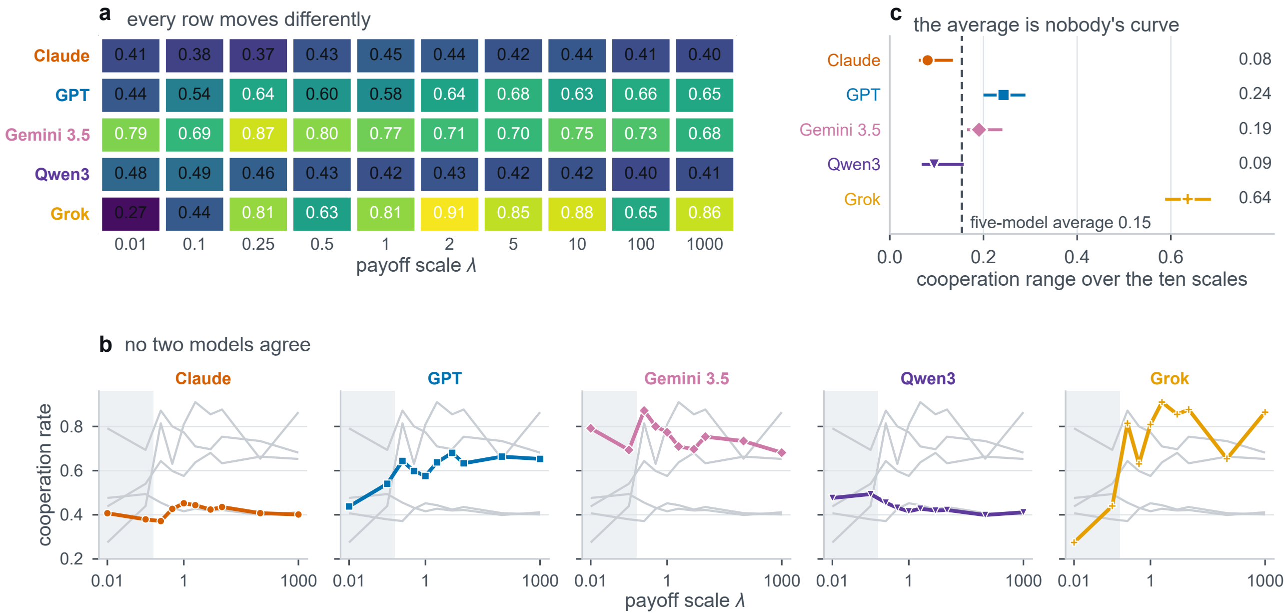
The largest departures are concentrated near the low-payoff end of the grid (table 1). Fractional multipliers often suppress cooperation relative to mid-ladder play, whereas changes from unit stakes to large stakes have no common direction across models. This boundary-aligned pattern is descriptive. The present design does not establish a discrete cognitive threshold, distinguish numerical magnitude from every feature of notation, or identify the mechanism behind the response.

Equally revealing is what the payoff scale does not change. While aggregate cooperation, opening dispositions, and mutual defection lock-in all respond dynamically to the stakes, the distribution of surplus between paired agents shows no detected association with payoff scale (table 1). As demonstrated in appendix B.9 and figure B.4, within-dyad inequality in symmetric interactions is an algebraic signature of coordination failure rather than an ethical preference. Scaling the stakes shifts the overall willingness of agents to cooperate, but we detect no corresponding association with the symmetric coordination-balance measure between paired clones.

3.2 The size of the effect depends on the prompt language

Prompt language is not a neutral transmission channel for the game. Both the baseline level of cooperation and its sensitivity to the stakes diverge substantially across prompt languages. Figure 3 shows the pooled language profiles, while the model-by-language levels and ranges appear in electronic supplementary table B.3.

One caveat attaches to the level comparison rather than to the scale result. The Arabic and Chinese templates differ from the other prompt stimuli in their objective wording. They state the four cells as penalties while phrasing the goal sentence as maximising a reward (§2.3). Because that wording is fixed within each language, the within-language comparisons



20,000 agent-games from 10,000 dyads, 200 dyads (400 agent-games) per model and payoff scale; shading in b marks the sub-unit regime; intervals in c are the 95% percentile whole-dyad intervals from the main statistical table.

Figure 2. Cooperation against the payoff scale across five frontier models. (a) Model-by-scale cooperation matrix: mean cooperation for each of the five models at each of the ten payoff scales $\lambda \in [0.01, 1000]$, with the value printed in each tile. (b) Small multiples per model: the focal model is drawn with its designated colour and marker, with the other four models shown behind in grey for context. Shaded bands mark the sub-unit regime ($\lambda \leq 0.1$). (c) Within-model cooperation range (maximum minus minimum rate across the ten scales) compared against the pooled five-model range (0.154, vertical dashed line). Confidence intervals are 95% percentile intervals from 1,500 whole-dyad bootstrap resamples.

Table 2. Persona effect below and above the sub-unit boundary, by model. Cooperation under a cooperative persona minus cooperation under a selfish one, at the two scales where every printed payoff is at most 1 ($\lambda \leq 0.1$) and at the eight where some payoff exceeds it. Intervals are 95% confidence intervals from the dyad bootstrap. A sign flip needs both regime intervals to exclude zero on opposite sides, so a supra-unit interval straddling zero means the effect is abolished rather than reversed.

	persona effect, with 95% CI				
Model	$\lambda \leq 0.1$	$\lambda \geq 0.25$	shift	95% CI	sign flip
Claude Haiku 4.5	-0.222 $[-0.258, -0.186]$	$+0.207$ $[+0.189, +0.226]$	0.429	$[0.389, 0.471]$	yes
GPT-5.4 Nano	-0.135 $[-0.179, -0.091]$	$+0.018$ $[-0.001, +0.037]$	0.153	$[0.105, 0.200]$	no
Gemini 3.5 Flash-Lite	-0.456 $[-0.507, -0.406]$	$+0.224$ $[+0.207, +0.242]$	0.680	$[0.625, 0.736]$	yes
Qwen3 235B-A22B	-0.911 $[-0.931, -0.893]$	-0.679 $[-0.694, -0.664]$	0.233	$[0.210, 0.258]$	no
Grok 4.20	-0.441 $[-0.485, -0.396]$	-0.303 $[-0.326, -0.279]$	0.138	$[0.090, 0.185]$	no

still establish sensitivity to payoff rescaling for the prompt as actually presented. However, differences in the magnitude or shape of scale sensitivity across languages cannot be attributed to language alone, because language is partially confounded with template wording. We therefore interpret the scale-by-language interaction as a prompt-stimulus interaction, not as evidence for a linguistic or cultural mechanism.

The central discovery is that scale sensitivity itself depends on the prompt language. The model-specific max-minus-min ranges over the ten scales vary by up to an order of magnitude within a model (electronic supplementary table B.3). Figure 3b provides the complementary supra-unit minus sub-unit contrast, and figure 3c shows that language differences remain after removing their mean levels. Across the full corpus, the observed sensitivity ranges from near-invariance to sweeping behavioural transformations spanning almost the entire probability scale.

Crucially, these linguistic biases do not point in a uniform direction across models: each model identifies a different language as its most cooperative medium and another as its most scale-sensitive. When multilingual evaluations pool across models, these opposing model-specific effects are attenuated and can obscure substantial model-by-language

Table 3. Rule agreement and rule distance against the payoff scale, by model. Left, the share of agent-games matched exactly by one canonical rule, below and above the sub-unit boundary, with the coefficient of a logistic regression of that indicator on $\log_{10}\lambda$; right, the mean number of rounds violating the nearest canonical rule at the smallest and largest scale, with the same trend by least squares. Both β are per decade of λ and their signs differ between models. Both cluster on the dyad, and neither block uses the learned classifier.

	exactly one rule fits (%)				rounds violating the nearest rule			
Model	$\lambda \leq 0.1$	$\lambda \geq 0.25$	β	p	λ=0.01	λ=10^3	β	p
Claude Haiku 4.5	17.9	3.5	−0.545	< 0.001	1.70	2.02	+0.073	< 0.001
GPT-5.4 Nano	9.8	7.6	−0.038	0.441	2.02	2.19	+0.038	0.052
Gemini 3.5 Flash-Lite	32.2	15.6	−0.242	< 0.001	0.48	1.36	+0.194	< 0.001
Qwen3 235B-A22B	72.1	53.6	−0.267	< 0.001	0.27	1.14	+0.202	< 0.001
Grok 4.20	44.6	17.1	−0.301	< 0.001	0.92	0.37	−0.118	< 0.001

heterogeneity. In reality, machine behaviour depends on the linguistic stimulus used in the prompt. Within this design, scale sensitivity is a joint property of the model and prompt language rather than a model-independent invariant. It does not identify a cultural mechanism.

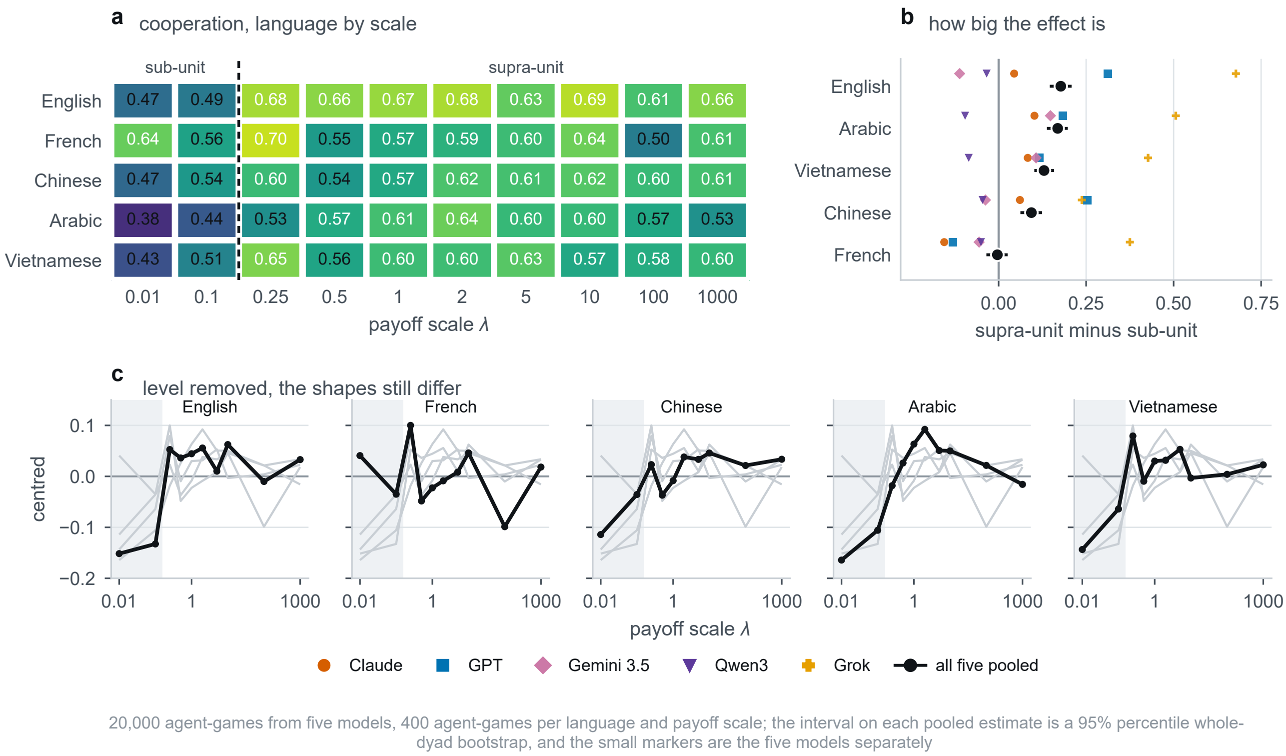


Figure 3. Cross-lingual scale sensitivity. (a) Cooperation pooled across the five main-text models for each prompt language and payoff scale; the dashed boundary separates the sub-unit and supra-unit regimes. (b) Language-specific difference in cooperation between supra-unit and sub-unit regimes; intervals show the pooled estimate and small markers the five models separately. (c) Cooperation curves after subtracting each language's mean, showing that differences in scale-response shape remain after level differences are removed.

3.3 Heterogeneity, and what pooling does to it

Averaging across models conceals the real phenomenon through mutual cancellation. The pooled response curve severely attenuates the true magnitude of scale sensitivity, shrinking the typical within-model movement substantially (figure 2c). Because pairwise correlations between model curves are predominantly negative (figure 2b), curves that move in opposite directions flatten each other out when summed. A pooled curve describes an artificial average agent that no real model resembles.

Underlying this cancellation is a fundamental divide in policy geometry: cooperation in these models is not a single Gaussian distribution, but a mixture over distinct policy regimes (appendix B.6). Across the corpus, unconditional boundary play is strongly overrepresented, with extreme atoms carrying far more weight than expected under uniform play. Crucially, the commitment to boundary play varies dramatically across models: Claude operates as an interior, mixed-policy player that rarely locks into extreme postures, whereas Qwen and Grok function as pure-policy switchers, oscillating between unconditional defection and unconditional cooperation. Pooling across models conflates how often a model chooses a pure strategy with which pure strategy it selects.

A formal variance decomposition confirms that model individuality dominates the game (table B.4, appendix B.3). Model identity and the model-by-language interaction account for the majority of the variance across cells. While the main effect of payoff scale is statistically detectable, its interaction with model identity carries three times as much variance, demonstrating that the behavioural impact of stakes is model-specific rather than shared universally across model endpoints.

In the five-model cell-mean variance decomposition, the shared payoff-scale main effect is largely carried by Grok. Removing Grok renders that particular main-effect test non-significant, even though every remaining model continues to move (appendix B.3). A complementary dyad-level leave-one-model-out test asks a different question, whether any pooled scale association remains. Every five-model subset retains a detectable association, including the subset without Grok (appendix D). The two analyses are therefore complementary: one tests a shared main effect after accounting for heterogeneity, whereas the other tests any residual pooled association.

3.4 The payoff scale gates the persona instruction

System prompts designed to steer agent behaviour do not operate independently of game stakes. The behavioural difference between cooperative and selfish persona instructions changes dramatically across the payoff ladder (figure 4, table 2). This modulation differs between the two sub-unit scales and the eight larger scales, where printed payoffs include values above one (§2.2). Across all five models, the persona effect changes across these two regimes.

In multiple frontier models, the persona instruction actively inverts below the unit threshold. In Gemini and Claude, the persona effect reverses sign: when stakes are fractional, these models cooperate *more* when explicitly commanded to be selfish than when told to be cooperative. In GPT, the persona effect is essentially extinguished at fractional scales rather than reversed, while the remaining models maintain their orientation. This result is a boundary-aligned descriptive effect. It does not identify why the prompt instruction changes direction at these scales.

The main grid does not separate numerical magnitude from all aspects of numeral formatting. We therefore treat this regime split as descriptive and do not attribute it to a specific representational mechanism.

A separate class of models inverts the persona instruction across the entire ladder (table 2). Qwen and Grok cooperate substantially more under the selfish persona than under the cooperative persona at every evaluated scale. These models show an opposite response to the persona wording. The data do not establish why they do so. This result underscores that role prompts cannot be assumed to steer multi-agent interactions in the intended direction without empirical verification across payoff contexts (appendix B.4).

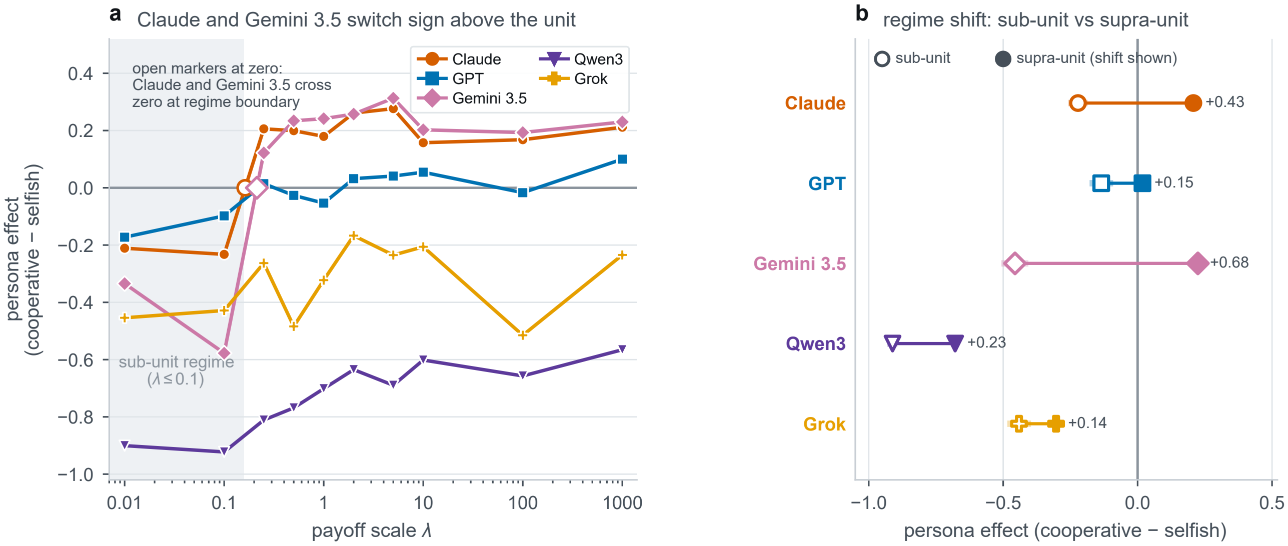
3.5 The scale changes what is played, not only how often

Two models displaying identical cooperation rates can be executing fundamentally different strategic policies. To understand the behavioural rules underlying raw cooperation rates, we read every agent trajectory against the four canonical memory-one strategies of evolutionary game theory (§2.4). What this read-out captures is behavioural resemblance rather than exact algorithmic identity: across the corpus, the majority of games match no textbook rule cleanly (table 3, appendix B.5).

As payoff stakes expand, play moves farther from the four canonical rules in most frontier models. Across the scale ladder, the proportion of games adhering strictly to a single canonical rule declines substantially, while unmatched play accumulates (table 3). This loss of canonical-rule conformity is pronounced in Claude, Grok, Qwen, and Gemini, while GPT maintains steady adherence. The read-out does not establish what rule generates unmatched play; it only shows that it is farther from the four canonical rules.

A label-free continuous metric confirms this behavioural drift: the number of rounds departing from the nearest canonical rule increases significantly with scale across most models (table 3). As stakes multiply, trajectories depart more often from the textbook templates. Only Grok reverses this trend, moving towards extreme pure-policy play at high stakes.

The qualitative composition of strategies also shifts dynamically, driving divergent realignments across models (figure 5). In GPT, expanding stakes prompts a steady migration from unconditional defection toward unconditional cooperation. In Gemini, the opposite shift occurs: unconditional cooperation yields ground to more Tit-for-Tat-like and Win-Stay-Lose-Shift-like attributions. Grok undergoes an outright reversal, swapping rampant defection at micro-stakes for overwhelming cooperation at high stakes.



20,000 agent-games from 10,000 dyads across five models and ten payoff scales; shading on a and open markers on b denote the sub-unit regime (lambda <= 0.1); thick bars on b are 95% bootstrap intervals; values right of markers are the supra minus sub-unit shift

Figure 4. Persona effect and sub-unit inversion against the payoff scale. The persona effect is the cooperation rate under the cooperative persona minus that under the selfish persona. (a) Difference in cooperation rate across scales λ, with the sub-unit regime ($\lambda \leq 0.1$) shaded where all five models cooperate more under the selfish prompt. (b) Regime means comparing sub-unit ($\lambda \leq 0.1$, open circles) with supra-unit ($\lambda \geq 0.25$, filled circles) regimes (table 2).

Critically, these strategy shifts reflect qualitative resemblances rather than textbook execution. Provenance auditing demonstrates that while unconditional cooperation and defection are often matched exactly, nearly all attributions of Tit-for-Tat and Win-Stay-Lose-Shift represent best-fit approximations to noisy, unmatched play (figure 5c, appendix B.5). When models appear to increase their Tit-for-Tat share, they are not adopting the classic reciprocal algorithm; they are generating complex, imperfect trajectories whose overall statistical contours happen to lean in that direction.

3.6 The effect is present before any strategic interaction

Rescaling stakes could alter how agents interpret ongoing gameplay, or it could act immediately on how they read the payoff matrix itself. The opening move cleanly isolates these possibilities: at round 1, the agent has seen only the matrix and prompt instructions, with zero interactive history.

For four of the five main-text models, the across-scale swing in opening-move cooperation matches or exceeds the range over rounds 2 to 10 (electronic supplementary table B.13). Qwen is the exception. Before observing a single action from its partner, these four models already show sensitivity to the nominal magnitude of the printed stakes.

The opening-move result localises part of the association to the interpretation of the initial matrix and prompt. It does not show that later interaction is irrelevant or identify an ex-ante mechanism.

Consistent with an early contribution of the printed matrix, round-by-round cooperation curves exhibit similar downward profiles across all ten rounds regardless of payoff scale (appendix B.8). Games played under micro-stakes and games played under massive stakes display similar downward slopes. This pattern is consistent with an end-of-game decline, but the design has no unknown-horizon condition and therefore does not identify backward induction as its mechanism.

3.7 Where the scale does act: the evolutionary baseline

While payoff scale is theoretically inert in normal-form games, evolutionary game theory provides a foundational framework where stakes legitimately matter, offering a rigorous theoretical benchmark for language model behaviour.

In deterministic replicator dynamics, scale is completely neutral. Multiplying the payoff matrix by a positive constant simply scales the velocity of the vector field: orbits follow identical geometric trajectories and all rest points remain invariant (appendix C.2, figure C.1).

In finite populations with stochastic imitation, however, payoff scale acts directly as selection intensity. Under pairwise Fermi imitation, scaling payoffs amplifies the competitive advantage of predatory exploitation over cooperation

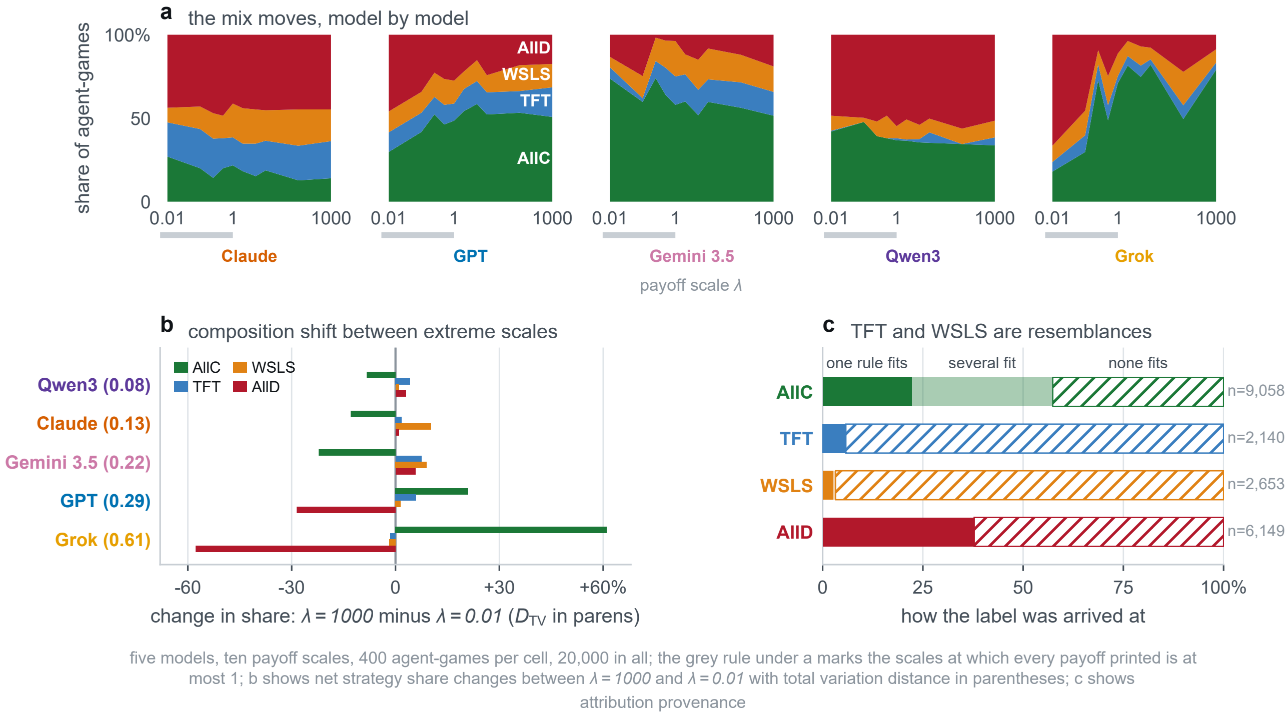


Figure 5. Strategy attribution across payoff scales, composition shift, and provenance breakdown. (a) Stacked area share of canonical memory-one strategy attributions (AllC, AllD, TFT, WSLS) across the ten payoff scales for each of the five models. (b) Net strategy share changes between extreme scales ($\lambda = 10^3$ minus $\lambda = 0.01$) across the four canonical rules with total variation distance D_{TV} in parentheses. (c) Strategy read-out provenance: exactly deduced (single canonical match), ambiguous (multiple matching rules), and unmatched (LSTM attribution on trajectories consistent with no textbook rule). Over the corpus, 94.6% of Tit-for-Tat and 97.1% of Win-Stay-Lose-Shift labels are attributions on unmatched play, establishing that labels indicate resemblance rather than faithful execution.

(appendix C). At micro-stakes, selection is weak and all four canonical rules coexist near genetic drift. As stakes expand, selection pressure intensifies: unconditional defectors ruthlessly invade cooperative rules until AllD achieves near-total fixation (figure 7a). Combining evolutionary dynamics with language-model evaluations provides a principled lens for multi-agent governance [40, 67].

At the pooled five-model level, the AllD label share moves in the opposite direction to the finite-population evolutionary baseline as λ increases (figure 7). This pooled reversal is driven primarily by GPT and Grok; per-model label trajectories remain heterogeneous. As payoffs grow into standard integer ranges, the pooled language-model corpus becomes more cooperative, whereas the evolutionary reference becomes more dominated by defection (figure 7b,c).

This direction is stable across the eighty evolutionary parameter settings reported in the appendix: the stationary AllD share is higher at the largest than at the smallest scale in every setting, and the rise is strictly monotone in 61 of 80 settings. None reproduces the direction of the pooled empirical trajectory. We therefore treat the comparison as a descriptive reference gap rather than as evidence for a particular mechanism. Evolutionary dynamics describe selection among rules, whereas the corpus describes fixed language-model agents.

The gap is a descriptive difference between selection among rules and fixed language-model agents. The language conditions also vary the stimulus, so this comparison does not identify a cultural or cognitive mechanism.

3.8 What the agents condition on

The evolutionary model rests on an explicit behavioural mechanism that can be evaluated directly in the data. In the classical account, cooperation survives invasion by defectors through *reciprocity*: conditioning one's choice on what the partner just did, retaliating against defection and rewarding cooperation. Tit-for-tat is the prototypical exemplar of this logic.

Because the previous round consists of an action pair (own action, partner action), the four conditional cooperation probabilities decompose cleanly into a base level and three orthogonal contrasts. *Reciprocity* represents the partner main effect: the difference in cooperation after a partner cooperates versus after a partner defects. *Persistence* represents the own

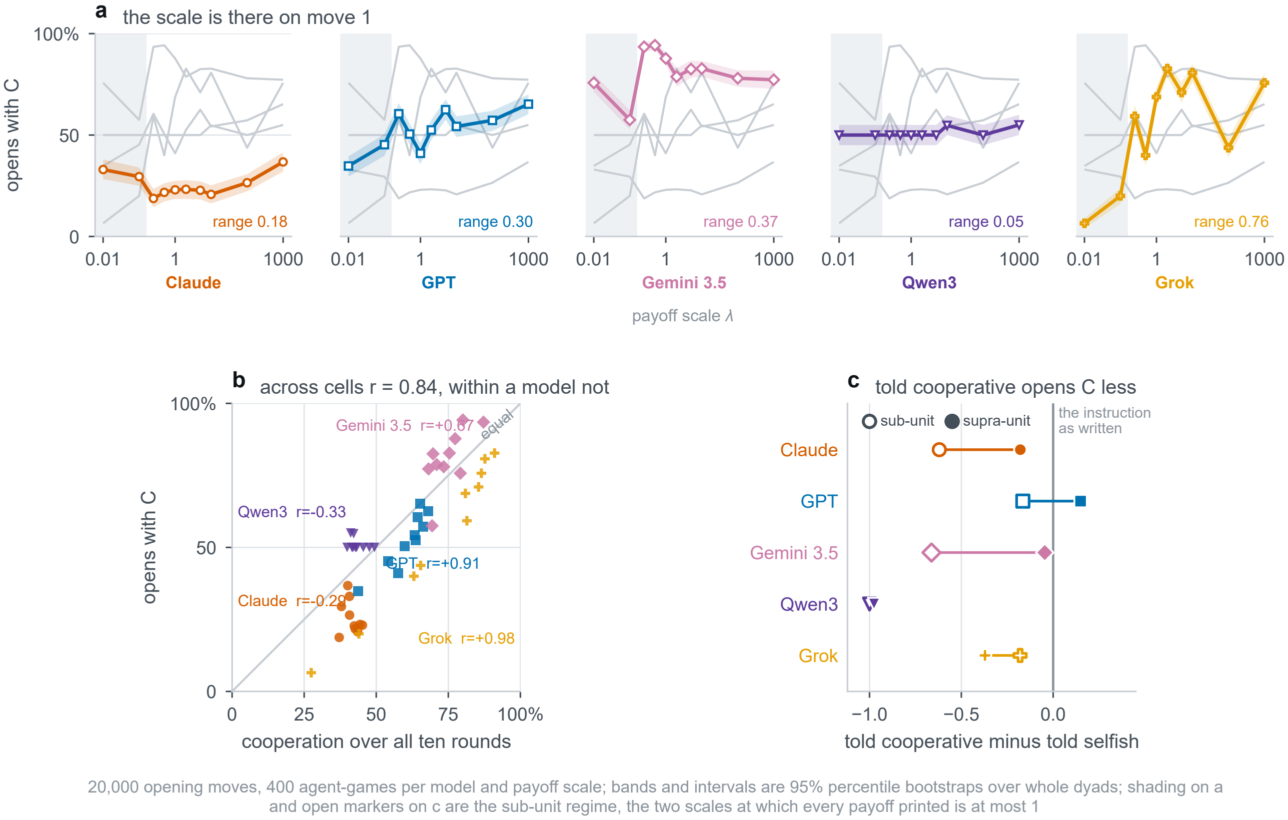


Figure 6. Opening-move sensitivity to payoff scale and persona. At round 1 the agent has observed only the matrix and instructions, with zero interaction history. (a) Opening-move cooperation across payoff scales for each model. (b) Opening-move cooperation against cooperation over all ten rounds across the 50 model-by-scale cells, with within-model correlations annotated. (c) Opening-move persona effect in the sub-unit and supra-unit regimes.

main effect: the tendency to repeat one's own previous action. *Matching* represents the interaction effect characteristic of Win-Stay-Lose-Shift. Textbook strategies occupy precise locations in this geometric space: Tit-for-Tat lies entirely on the reciprocity axis, Win-Stay-Lose-Shift on the matching axis, and unconditional rules at the origin (figure 8a).

Decomposing agent play reveals that own-action conditioning exceeds partner conditioning at the model level (figure 8). Across all six models, the persistence contrast is larger than the reciprocity contrast, although the margin is modest for Claude and substantially larger for the other models. Reciprocity remains positive in four models and is negative in Gemini 3.1 and Qwen. An unsupervised principal component analysis on raw transition vectors independently recovers this identical geometry without any prior knowledge of the contrasts (figure 8c).

The contrast estimates are precise at the dyad level. Whole-dyad bootstrap intervals for reciprocity are positive for Claude, GPT, Gemini 3.5, and Grok, and negative for Gemini 3.1 and Qwen; the corresponding persistence intervals are positive for all six models (electronic supplementary table D.1). Thus the headline comparison is one of relative strength, not a claim that partner information is absent.

This finding narrows one candidate explanation for the divergence from the evolutionary baseline. In these data, agents condition more strongly on their own preceding moves than on their partner's behaviour. The contrast does not show that partner information is absent, nor does it identify the causal basis of cooperation.

A technical caution governs this decomposition. The naive difference between disagreement states, $p_{CD} - p_{DC}$, is frequently mistaken for a reciprocity metric in the literature. In reality, it equals persistence minus reciprocity identically, confounding the two main effects and falsely indicating anti-reciprocal behaviour where the true partner effect is positive.

4. Discussion

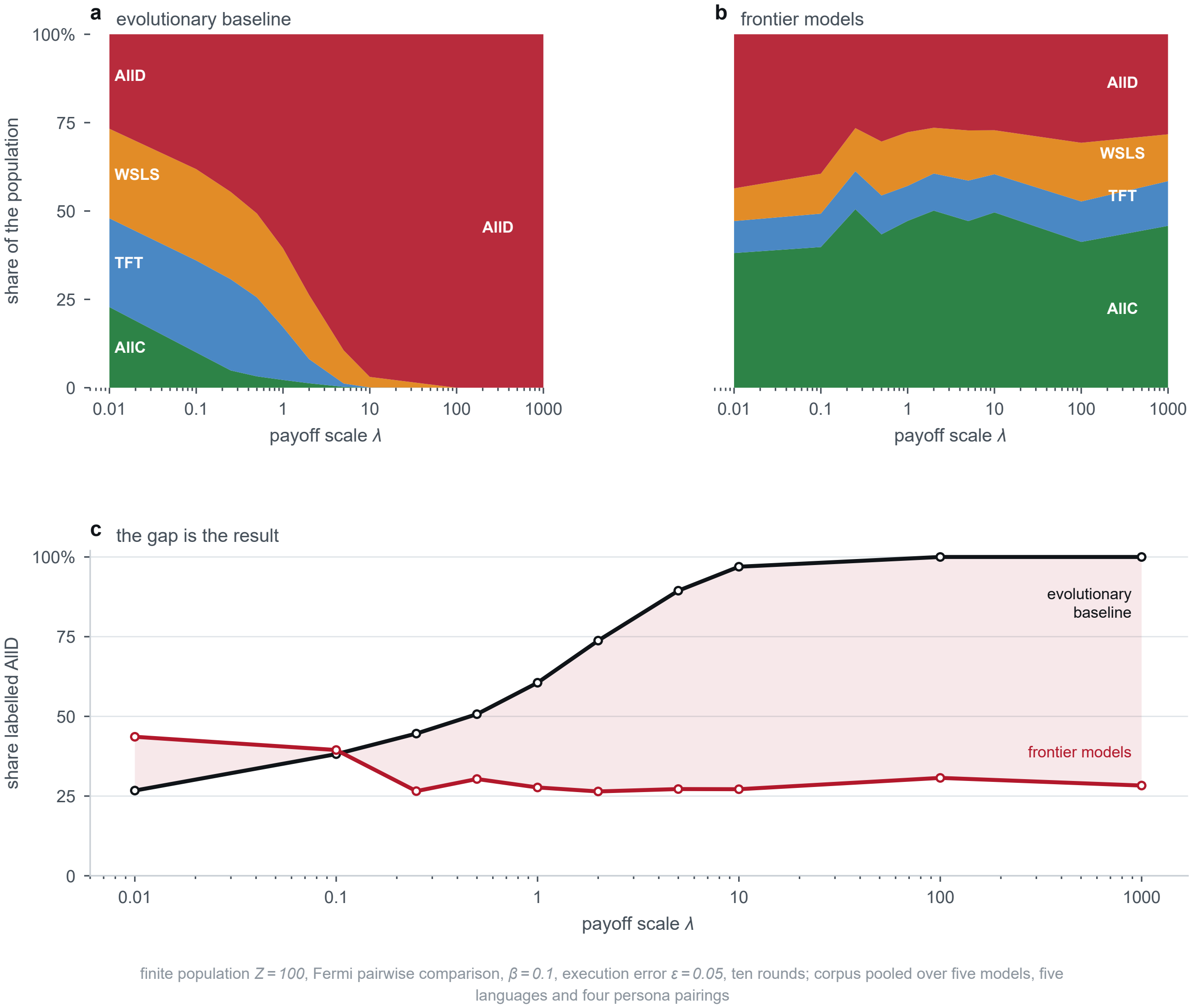


Figure 7. The evolutionary baseline and the corpus move in opposite directions. (a) The baseline's stationary distribution over the four canonical rules across the ten payoff scales, drawn as a stacked band. (b) The corpus's strategy mix on the same axis, drawn identically. (c) The AllD share alone, both accounts on one axis, with the gap shaded. Baseline at $Z = 100$, $\beta = 0.1$, $\epsilon = 0.05$, ten rounds; the corpus pooled over five models, five languages and four persona pairings.

4.1 Payoff-scale invariance is violated

The measurement reported here is unusual in having a null that is a theorem. That property lets a movement in cooperation be read as a departure from an invariance rather than as the difference between two conditions whose expected gap was never specified. The predicted difference between a matrix and the same matrix with larger cells is not small, or contested. It is zero.

All five models depart from it, and none is close to invariant. What makes the departures hard to set aside is not their size but their structure: they are present on the opening move, before any history exists, their size depends on the language the game is posed in, they gate the persona instruction at a boundary of magnitude, and they change what the transcripts look like as sequences. Sampling variability alone is difficult to reconcile with a pattern of that shape. The models appear to be responding to something in the matrix other than the game it encodes, though this design does not identify what.

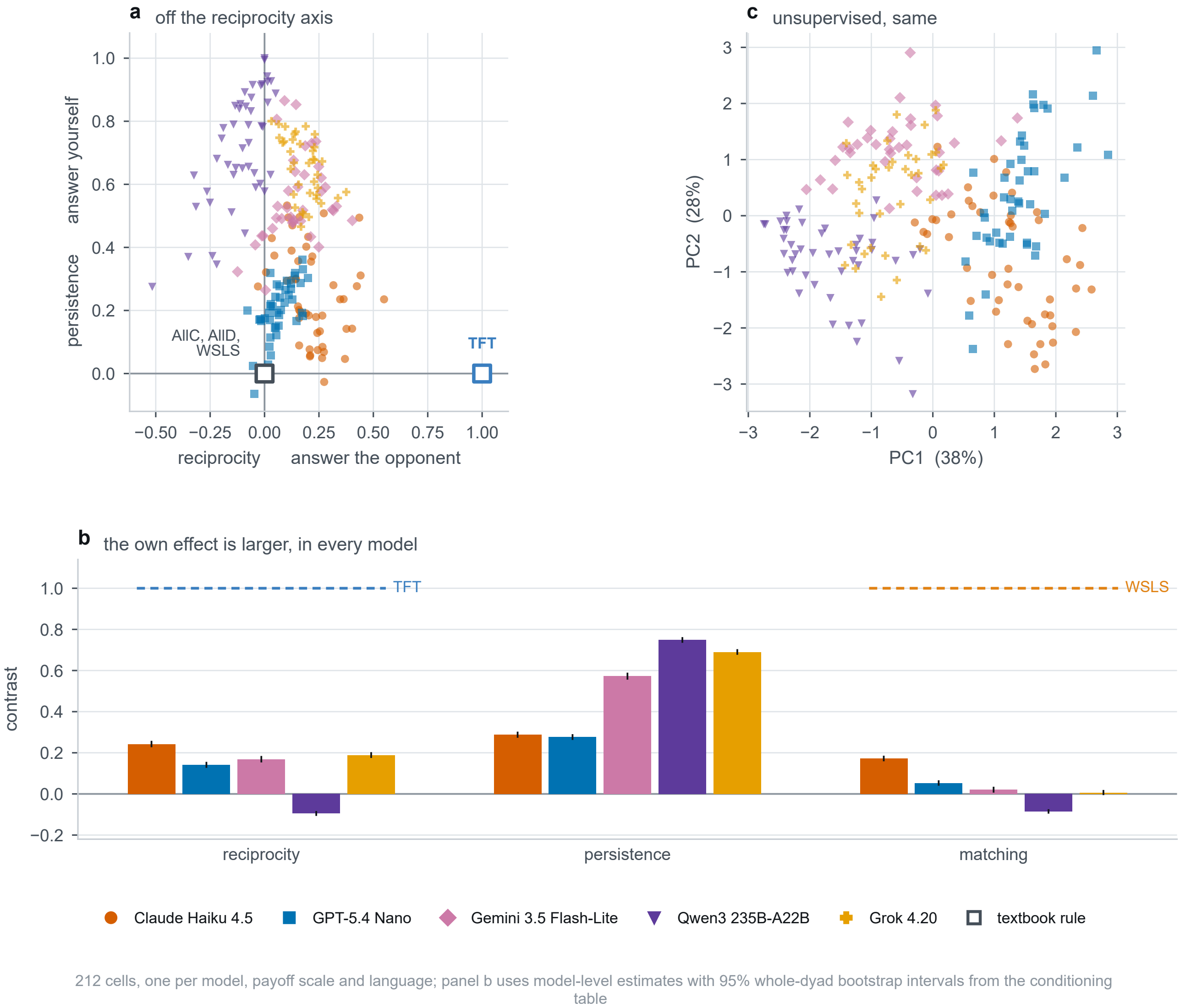


Figure 8. Own-action conditioning exceeds partner-action conditioning at the model level. (a) Cells in the reciprocity by persistence plane, with the canonical rules at their exact coordinates; AllC, AllD and win-stay lose-shift all have both main effects zero and so coincide at the origin. (b) The three contrasts per model, against the value each canonical rule would give, with 95% percentile intervals from whole-dyad bootstrap resampling. (c) The first two principal components of the raw five-vector, which is given no knowledge of the contrasts and recovers the same geometry. Panels (a) and (c) use model-by-scale-by-language cells; a predecessor observed fewer than thirty times yields no estimate and the cell is dropped, leaving 212 of 250.

4.2 Heterogeneity, and why pooled designs miss it

The pooling result deserves emphasis, because it is the mechanism by which an effect of this size could have gone unnoticed. The cause is arithmetic rather than statistical: the shapes disagree, and several pairs are anticorrelated. Heterogeneous effects that cancel are invisible to pooled designs by construction.

In this panel the cancellation is partial rather than total. A main effect of the payoff scale does survive the averaging, and it is significant, but it carries a third of the variance its interaction with the model carries, and the pooled range understates the typical model by a factor of 1.62: a statement about an average model that nobody deploys. A conclusion that changes when one model joins or leaves a panel is not a conclusion about language models. In the cell-mean shared-main-effect test, removing Grok makes the scale term non-significant; the complementary dyad-level leave-one-model-out association remains detectable. We therefore do not defend the pooled average as a single model-level quantity.

The pooled estimate is therefore not a weaker version of the per-model result but a different quantity, and it is not a question of power either: more dyads or more models only tighten the interval around an average of curves that point in different directions. The remedy is a per-model analysis, not a larger sample. The same logic runs one level deeper, since the effect depends on the prompt language as well as on the model, so a per-model number pooled over languages is the same trick played on a smaller stage. The heterogeneity is not a nuisance to be averaged away. It is the finding.

4.3 Implications for benchmarking and governance

A cooperation rate published for a language model is currently understood as a property of the model. On this evidence it is a joint property of the model, the units the matrix was written in, and the language the game was posed in. The units are the awkward member of that list, because they are the one nobody is asked to record: framings get described, personas get quoted, prompts get reproduced in appendices, while the payoff scale is fixed once, silently, on the authority of a theorem saying the choice is empty. Two laboratories running the same protocol on the same model, with matrices differing only by a constant rescaling factor, could report radically conflicting cooperation rates spanning more than half the probability scale.

We would suggest that a scale-sensitivity measurement accompany a published cooperation rate, in the way an error bar does, and be measured per model, most of all where such rates leave the benchmark and become the parameters of a population model. The same argument applies to safety auditing. An audit that runs in English at one fixed payoff matrix tests a single point on a much larger behavioural surface, and the two cheapest stress dimensions to add are the two measured here: a sweep over the payoff scale, and a parallel run in the prompt's non-English variants. Our central claim is complementary to a parallel line of work asking how external *mechanisms* change cooperation, through contracts and mediation [68], moral framing [69] or pre-play communication [70]. Payoff scale joins that list as an intervention on the numerical structure of the game rather than on the institutions around it. The behavioural profiles it produces are also the input that models of hybrid human-machine populations consume [7], and a profile that depends on the experimenter's units is not the quantity such a model needs.

4.4 Interpretation and possible mechanisms

Two observations constrain what an explanation must account for. The boundary gating the persona instruction appears to be one of magnitude rather than of notation, a separation only as strong as the single scale that supports it (§3.4); and the payoff scale moves how closely play tracks any canonical memory-one rule at all, and shifts the mix of labels with it. The scale is not turning a single dial marked cooperativeness; it is changing the shape of the trajectory.

Several accounts are consistent with both. The printed magnitudes may be read as stakes, which is exactly what a rescaling does not change; numerals of different magnitudes are tokenised differently and are met at very different frequencies in training text, so a representation that changes with the digit string could change what follows from it with no notion of stakes involved [71]; training text may associate large quantities with caution; and sub-unit payoffs may fall outside the range in which payoff-like numbers are ordinarily met. We can order none of these. The design measures departures from an invariance; it does not manipulate tokenisation, vary notation at fixed magnitude beyond the one contrast above, or move the framing that supplies the stakes. Nothing here should be read as identifying a mechanism.

There is a constructive reading as well as a cautionary one. A programme is under way to widen the utility function beyond the economic consequences of an action, so that the language describing the action enters it directly, with the replicator dynamics named as the destination [57]. Our result sits one level below that programme rather than beside it. The evidence for language-based preferences comes from manipulations that could legitimately have mattered, since a label carries connotation, whereas a positive rescaling carries none and is inert by theorem, and the behaviour still moves. A utility that mixes a payoff with a score computed from language is no longer scale-free in the payoff argument unless the two terms are rescaled together, so before such an object is carried into an evolutionary dynamic its invariance class has to be stated. What these data say is that at least one class of agents does not behave as though the payoff argument were scale-free, and that anything built to describe them will have to account for it.

4.5 Limitations and future work

This design does not identify the mechanism inside the model that produces the departures it measures across a broad but finite grid. Six models cannot support claims about model families or about parameter count, and what generalises here is that the shapes differ rather than what any one of them is. The corpus is one game, one prompt template family and one horizon, so whether the same curves appear in other dilemmas or in longer play is untested.

A sixth model, Gemini 3.1 Flash-Lite, was run on the identical grid and is reported in full in appendix B.7 rather than here, because its identifier names an unpinned preview endpoint and what was served under that name during collection cannot be recovered (§2.3). Two consequences are worth stating rather than leaving to be discovered. First, it is the only model of the six whose strategy trend is positive, so the four-of-five statement in §3.5 reads, over the whole corpus, as four models of six moving one way, one showing no trend and one moving the other; the second form is the one the corpus supports. Second, including it changes no conclusion here: the all-six pooled range, attenuation, shape correlations and variance decomposition are in appendix D.1, and every sign, significance and ordering in them agrees with the five-model versions reported above.

One feature of the stimuli bounds the language comparison. Two of the five templates state the payoffs as penalties while phrasing the goal as maximising a reward, and we reproduced rather than repaired them so that the corpus matches the published protocol. The wording is fixed within each language across the ten scales, so it does not invalidate the within-language rescaling comparison. It may nevertheless interact with numerical magnitude; consequently, cross-language differences in scale sensitivity cannot be attributed to language alone.

Our reference for calling a departure large is the sampling variability of these data, a bootstrap over dyads and a permutation null. A targeted independent repeat of Gemini 3.5 at four scales is reported in the appendix; it is useful evidence about repeatability in that arm but is not a test-retest floor for the full corpus. Clustering on the dyad addresses the dependence between the two agents of a game and between the rounds within an agent-game, but it does not turn the design into a mixed-effects model with random intercepts for replicate and game, which would be the more complete treatment of a nested design.

The strategy read-out carries a further caveat, which is why its provenance is reported wherever a label appears (§2.4). A strategy is a decision rule and not a decision, so it is never observed directly, and even where the resident rule is fixed by construction, separating the memory-one rules from observed play has been shown to need a long run of observations [38]. Our trajectories are ten rounds long. The claim that play becomes less rule-governed rests on exact matching and no learned component. The classifier was validated to 20% execution noise, which is two rounds in ten, and in two of the five models play sits about that far from the nearest canonical rule on average, so the read-out is working at the edge of the range it was validated on. The compositional claim uses labels the classifier supplies wherever the rules leave the answer open, is unevenly anchored across the four rules, and is the weaker of the two. Any reading of it as evidence that these models implement Tit-for-Tat would be unsupported by our data. Richer strategy vocabularies, including zero-determinant and extortionate play [26, 39] and mixture estimators over longer horizons [28, 37], are the natural extension.

Finally, the evolutionary baseline maps the stochasticity of a language model onto a symmetric per-action error rate, which is a clean theoretical object but a coarse abstraction: prompt sensitivity, end-game reasoning and the heterogeneity documented above are not well captured by symmetric flips. The baseline should be read as a reference distribution rather than as a fitted model, and the noise calibration reported in appendix C.8 is a first step rather than an estimate.

5. Conclusion

A positive rescaling of a payoff matrix is the rare manipulation whose predicted effect is not merely small but exactly zero. Five frontier language models were played across ten such rescalings, and every one of them moved. The movement is not a uniform bias a calibration could absorb: its shape differs by model and its size by prompt language, so most of it disappears from any analysis that averages models, and what survives the averaging depends on which models happen to be in the average. That is how a departure of this size can sit inside a literature that has not reported it. Read against an evolutionary baseline computed on the same matrices, the pooled corpus's AllD share moves in the opposite direction as the payoffs grow. This measures rather than explains the distance between what the game rewards and what these agents do.

What follows is a question about design rather than a verdict on any model. A cooperation rate measured on a language model is reported as though the matrix behind it were an arbitrary and therefore harmless choice. It is arbitrary, and it is not harmless. The same argument extends to every invariance the theory supplies, from shifts of the payoffs to relabelling of the actions, and each is an equally cheap test of whether an agent is playing the game it was shown.

Ethics. This work did not involve human participants, animal subjects or fieldwork, and did not require ethical approval from a human subject or animal welfare committee; all experiments are computational simulations of interactions between large language model agents.

Data Accessibility. The corpus of game transcripts, the prompt templates, the trained strategy classifier and the complete analysis pipeline that produces every figure, table and quoted number in this article are available to the editors, reviewers and readers at https://github.com/trungkiet2005/prisoner-dilemma-llm. The transcripts are released under the Creative Commons Attribution 4.0 International licence and the analysis code under the MIT licence. A versioned archive and DOI will be added on publication.

Declaration of AI Use. Large language models are the *object* of study in this article: the corpus consists of transcripts of games played by the six models named in §2.3, collected through an automated pipeline. Generative AI was additionally used as a coding assistant for parts of the analysis and figure code, and for language editing of the manuscript. All analyses were designed, run and checked by the authors; tables and figures are generated from the archived pipeline, and an automated verification ledger checks the manually quoted headline numerical claims against the released data. The authors take full responsibility for the content of this article.

Authors' Contributions. T.A.H.: conceptualisation, methodology, supervision, writing (original draft), writing (review and editing). T.-K.H.: conceptualisation, software, formal analysis, visualisation, writing (original draft). D.-S.D.-M.: software, data curation, formal

analysis, writing (original draft). T.-B.C., P.-H.L., H.-D.N., P.-Q.N.-L., M.-L.N.-V., H.-P.P., P.-H.P., T.-K.T., C.-N.T., H.T. and G.-T.T.-L.: investigation, data curation, validation, writing (review and editing). A.B.: methodology, resources, writing (review and editing). L.H.T.: conceptualisation, methodology, supervision, project administration, writing (review and editing). T.A.H., T.-K.H. and D.-S.D.-M. contributed equally to this work. All authors gave final approval for publication and agree to be held accountable for the work performed therein.

Competing Interests. We declare we have no competing interests.

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